

streams in java:

1. a stream is nothing but flow of information.
2. package: java.io
3. input -----> program -----> output
4. input -----> program -----> output -----> output.txt
5. files or data bases
6. files -----> unformatted data will be there
7. database --> formatted data (mobile--> only mobile numbers)

Dynamic Data Reading

File Processing

Database connectivity

All the streams are divided into two types

1. Byte Oriented Streams -----> bytes
2. Character Oriented Streams ----> character

Reading of data dynamically from the user

1. BufferedReader -----> char and String
2. Scanner -----> read all primitives
(byte,short,int,long,float,double,char,boolean)
3. Console -----> read Strings and Passwords

Eg:

```
import java.io.InputStreamReader;
import java.io.BufferedReader;
class Test
{
    public static void main(String[] args) throws Exception
    {
        BufferedReader br = new BufferedReader(new
        InputStreamReader(System.in));
        char ch = (char)br.read();//read function accepts an int value --->
        char
        System.out.println("The given character:"+ch);
    }
}
```

D:\>javac Test.java

D:\>java Test

A

The given character:A

D:\>java Test

p

The given character:p

Eg:

```
import java.io.InputStreamReader;
import java.io.BufferedReader;
class Test
{
    public static void main(String[] args) throws Exception
    {
        BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));
        String s = br.readLine();
        System.out.println("The given String:"+s);
    }
}
```

D:\>javac Test.java

D:\>java Test

Praakash Babu

The given String:Praakash Babu

Eg:

```
import java.util.Scanner;
class Test
{
    public static void main(String[] args) throws Exception
    {
        Scanner obj = new Scanner(System.in);
        int x = obj.nextInt();
        System.out.println("Given Integer:"+x);
    }
}
```

D:\>java Test

145

Given Integer:145

Eg:

```
import java.util.Scanner;
class Test
{
    public static void main(String[] args) throws Exception
    {
        Scanner obj = new Scanner(System.in);
        double x = obj.nextDouble();
        System.out.println("Given Double:"+x);
    }
}
```

D:\>javac Test.java

D:\>java Test

1.23456

Given Double:1.23456

Eg:

```
import java.util.Scanner;
```

```
class Test
```

```
{
```

```
    public static void main(String[] args) throws Exception
```

```
    {
```

```
        Scanner obj = new Scanner(System.in);
```

```
        char x = obj.nextLine().charAt(0);
```

```
        System.out.println("Given character:"+x);
```

```
    }
```

```
}
```

D:\>javac Test.java

D:\>java Test

a

Given character:a

D:\>java Test

b

Given character:b

Eg:

```
import java.util.Scanner;
```

```
class Test
```

```
{
```

```
    public static void main(String[] args) throws Exception
```

```
    {
```

```
        Scanner obj = new Scanner(System.in);
```

```
        boolean b = obj.nextBoolean();
```

```
        System.out.println("Given boolean:"+b);
```

```
    }
```

```
}
```

D:\>javac Test.java

D:\>java Test

true

Given boolean:true

D:\>java Test

false

Given boolean:false

Eg:

```
import java.util.Scanner;
class Test
{
    public static void main(String[] args) throws Exception
    {
        Scanner obj = new Scanner(System.in);
        String s = obj.next();
        System.out.println("Given String:"+s);
    }
}
```

D:\>javac Test.java

D:\>java Test

Prakash

Given String:Prakash

D:\>java Test

Prakash Babu

Given String:Prakash

Eg:

```
import java.util.Scanner;
class Test
{
    public static void main(String[] args) throws Exception
    {
        Scanner obj = new Scanner(System.in);
        String s = obj.nextLine();
        System.out.println("Given String:"+s);
    }
}
```

D:\>javac Test.java

D:\>java Test

Prakash

Given String:Prakash

D:\>java Test

Prakash Babu

Given String:Prakash Babu

Console:

Eg:

```
import java.io.Console;
class Test
{
```

```

    public static void main(String[] args) throws Exception
    {
        Console c = System.console();
        String uname = c.readLine("Enter username:");
        System.out.println("username="+uname);
        char[] ch = c.readPassword("Enter password:");
        String password=new String(ch);
        System.out.println("password="+password);
    }
}

```

D:\>javac Test.java

```

D:\>java Test
Enter username:admin
username=admin
Enter password:
password=admin@123

```

Writing output to the user

```

-----
1. System.out.print();
2. System.out.println();
3. System.out.printf();

```

Reading & Writing Byte Oriented Streams

=====

Eg1:

```

import java.io.FileOutputStream;
class Test
{
    public static void main(String[] args) throws Exception
    {
        FileOutputStream fos = new FileOutputStream("abc1.txt");
        String data = "welcome to java";
        byte[] b = data.getBytes();
        fos.write(b);
        fos.close();
    }
}

```

Eg2:

```

import java.io.FileInputStream;
class Test
{
    public static void main(String[] args) throws Exception
    {
        FileInputStream fis = new FileInputStream("Test.java");
    }
}

```

```

        int size = fis.available();
        byte[] b = new byte[size];
        fis.read(b);
        String data = new String(b);
        System.out.println(data);
        fis.close();
    }
}

```

Eg3:

```

import java.io.FileInputStream;
import java.io.FileOutputStream;
class Test
{
    public static void main(String[] args) throws Exception
    {
        FileInputStream fis = new FileInputStream("abc1.txt");
        FileOutputStream fos = new FileOutputStream("abc2.txt");
        int size = fis.available();
        byte[] b = new byte[size];
        fis.read(b);
        fos.write(b);
        fos.close();
        fis.close();
    }
}

```

Eg4:

```

import java.io.FileInputStream;
import java.io.FileOutputStream;
class Test
{
    public static void main(String[] args) throws Exception
    {
        FileInputStream fis = new FileInputStream("abc1.txt");
        FileOutputStream fos = new FileOutputStream("abc3.txt");
        int size = fis.available();
        byte[] b = new byte[size/2];
        fis.read(b);
        fos.write(b);
        fos.close();
        fis.close();
    }
}

```

Character Oriented Streams

Readers: FileReader,BufferedReader

Writers: FileWriter,BufferedWriter,PrintWriter

FileWriter:

Constructors:

FileWriter fw = new FileWriter(String fname);
FileWriter fw = new FileWriter(String fname,boolean);

Methods:

write(int) ----> ASCII value
write(char[]) -> writes char[] into file
write(String)--> write a string to the file
close() -----> close the file

Eg:

```
import java.io.FileWriter;
class Test
{
    public static void main(String[] args) throws Exception
    {
        FileWriter fw = new FileWriter("abc4.txt");
        fw.write(97);//a
        fw.write("\n");
        char[] ch = {'b','c','d'};
        fw.write(ch);//bcd
        fw.write("\n");
        fw.write("Java Porgramming");//Java Porgramming
        fw.write("\n");
        fw.close();
    }
}
```

Eg:

```
import java.io.FileWriter;
class Test
{
    public static void main(String[] args) throws Exception
    {
        FileWriter fw = new FileWriter("abc4.txt",true);//apend
        fw.write("C Porgramming");
        fw.write("\n");
        fw.write("C++ Porgramming");
        fw.write("\n");
        fw.write("JAVA Porgramming");
    }
}
```

```

        fw.write("\n");
        fw.close();
    }
}

```

Problem with FileWriter: \n ----> new line character in java

BufferedWriter:

Constructor

```
BufferedWriter bw = new BufferedWriter(new FileWriter(String fname));
```

Methods

1. write(int);
2. write(char[]);
3. write(String)
4. close();
5. newLine();

Eg:

```

import java.io.FileWriter;
import java.io.BufferedWriter;
class Test
{
    public static void main(String[] args) throws Exception
    {
        BufferedWriter bw = new BufferedWriter(new FileWriter("abc5.txt"));
        bw.write("Chiranjeevi");
        bw.newLine();
        bw.write("Venkatesh");
        bw.newLine();
        bw.write("Balakrishna");
        bw.newLine();
        bw.write("Nagarjuna");
        bw.close();
    }
}

```

Problem with BufferedWriter -----> 1K -----> 1K+1K newLine()

PrintWriter:

Constructors:

```
PrintWriter pw = new PrintWriter(String fname);
```

methods:

1. println(int)
2. println(byte)
3. println(short)
4. println(long)
5. println(float)
6. println(double)
7. println(char)
8. println(boolean)
9. println(String)
10. close()

Eg:

```
import java.io.PrintWriter;
class Test
{
    public static void main(String[] args) throws Exception
    {
        PrintWriter pw = new PrintWriter("abc6.txt");
        pw.println(123);
        pw.println(12.3456);
        pw.println('a');
        pw.println(true);
        pw.println("prakash");
        pw.close();
    }
}
```

Readers: FileReader,BufferedReader

+++++

FileReader:

Constructors

```
FileReader fr = new FileReader(String fname);
```

Methods:

1. read() ----> a single char in ascii format
2. close()

Eg:

```
import java.io.FileReader;
class Test
{
```

```

    public static void main(String[] args) throws Exception
    {
        FileReader fr = new FileReader("abc.txt");
        int n = fr.read();
        while(n!=-1)
        {
            System.out.print((char)n);
            n=fr.read();
        }
        fr.close();
    }
}

```

BufferedReader

Constructor

```
BufferedReader br = new BufferedReader(new FileReader(String fname));
```

Methods:

1. int read()
2. String readLine()
3. close()

Eg:

```

import java.io.FileReader;
import java.io.BufferedReader;
class Test
{
    public static void main(String[] args) throws Exception
    {
        BufferedReader br = new BufferedReader(new FileReader("abc.txt"));
        String line = br.readLine();
        while(line!=null)
        {
            System.out.println(line);
            line=br.readLine();
        }
        br.close();
    }
}

```

File Class:

```
File f = new File(String fname);
```

Eg:

```
import java.io.File;
```

```

class Test
{
    public static void main(String[] args) throws Exception
    {
        File f = new File("abc.txt");
        System.out.println(f.exists()); // True
    }
}

```

Eg:

```

import java.io.File;
class Test
{
    public static void main(String[] args) throws Exception
    {
        File f = new File("xyz.txt");
        System.out.println(f.exists()); // False
        // f.createNewFile();
        // System.out.println(f.exists()); // True
    }
}

```

Eg:

```

import java.io.File;
class Test
{
    public static void main(String[] args) throws Exception
    {
        File f = new File("king");
        System.out.println(f.exists()); // False
        // f.mkdir();
        // System.out.println(f.exists()); // true
    }
}

```

Eg:

```

import java.io.File;
class Test
{
    public static void main(String[] args) throws Exception
    {
        File f = new File("abc.txt");
        System.out.println(f.length()); // 75
    }
}

```

Eg:

```
import java.io.File;
class Test
{
    public static void main(String[] args) throws Exception
    {
        File f1 = new File("abc.txt");
        File f2 = new File("king");
        System.out.println(f1.isFile());//true
        System.out.println(f2.isFile());//false
        System.out.println(f1.isDirectory());//false
        System.out.println(f2.isDirectory());//true
    }
}
```